

UNIVERSITÄT PADERBORN

Assignment 3 Report

Data Science for Software Engineering

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Chapter 1

Introduction

1.1 Introduction

Issue tracking systems such as Jira contain useful software engineering discussions about bugs, improvements, features and design decisions. Prior work has shown that bug reports provide important information for developers, although the quality and completeness of this information can vary [2]. In this task, we focus on Apache Tika issues from the Apache Jira issue tracker [1]. The overall goal is to explore the *topics* of design issues using topic modeling (LDA and BERTopic) and to study their characteristics and their co-occurrence with three types of design decisions: *existence*, *property* and *executive*.

Chapter 2

Study Design

We used the Tika sheet from the provided Excel file because Apache Tika is our selected project. The sheet contained 211 issue IDs and three design-decision labels: existence, property and executive.

Using Python, we downloaded the corresponding issue data from Apache Jira. The downloaded fields included summary, description, issue type, status, resolution, priority, labels, components, dates, parent issue, number of comments and number of attachments.

For preprocessing, we combined each issue summary and description into one text field. The text was cleaned by lowercasing, removing URLs and special characters, splitting camel-case words, removing stop words, lemmatizing tokens and removing very short tokens.

For Week 2, Latent Dirichlet Allocation (LDA) topic modeling was applied in two iterations using the Gensim library [3]. In both iterations the model parameters were $\alpha = 0.01$, $\beta = 0.01$, 20 training passes, 400 iterations per pass, and a fixed random seed of 50. The low symmetric α encourages sparse topic distributions so that each issue is strongly associated with one dominant topic. Iteration 1 used a fixed number of 10 topics. To find the optimal number of topics, C_V coherence scores were computed for $k = 1$ to $k = 10$, and the k with the highest coherence was selected for Iteration 2. For Iteration 2, each token was replaced with its ontology class label (COMPONENT, CONNECTOR, DATA, SOLUTION, TECHNOLOGY, QUALITY_ATTRIBUTE) where it appeared in the ontology, and LDA was re-run with the optimal k .

The second topic model, BERTopic [4], was run on the same preprocessed text using `all-MiniLM-L6-v2` sentence embeddings [5]. Because the small corpus is dominated by one dense cluster, the default HDBSCAN clustering produced only two topics; we therefore used KMeans ($k = 7$, matching the LDA optimum) so that BERTopic yields a comparable, balanced set of topics with every issue assigned.

For Week 3 we analysed the issues of each topic. Numeric issue characteristics (comments, attachments, description length, token count) were compared across topics with Kruskal–Wallis tests and Dunn post-hoc tests; issue type was compared with χ^2 tests. Co-occurrences between LDA topics, BERTopics and the three decision types were tested with χ^2 tests, standardized residuals, and Fisher exact tests with Benjamini–Hochberg correction.

Chapter 3

Results

3.1 Week 1: Issue Data Download and Vocabulary Creation

A total of 211 Apache Tika issues were processed. All issues were successfully downloaded from Jira using the provided issue list. For each issue, the summary, description, issue type, status, priority, parent-issue information, number of comments and number of attachments were collected. As expected, most issues have no parent: only 10 of the 211 issues (4.7%) reference a parent issue.

After downloading, the issue summary and description were combined and preprocessed. After preprocessing, no issue had zero tokens; the average number of tokens per issue was 69.78. The vocabulary contained 14,724 total tokens and 2,706 unique tokens. The most frequent tokens included `parser`, `file`, `metadata`, `type`, `text`, `content`, `test`, `code`, `document`, `version`, `pdf`, `user`, `server`, `stream`, `data`, `parse`, `class`, `java`, `vulnerability` and `cve`. A document-term matrix of shape 211×2706 was created and used as input for LDA (Figure 3.1).

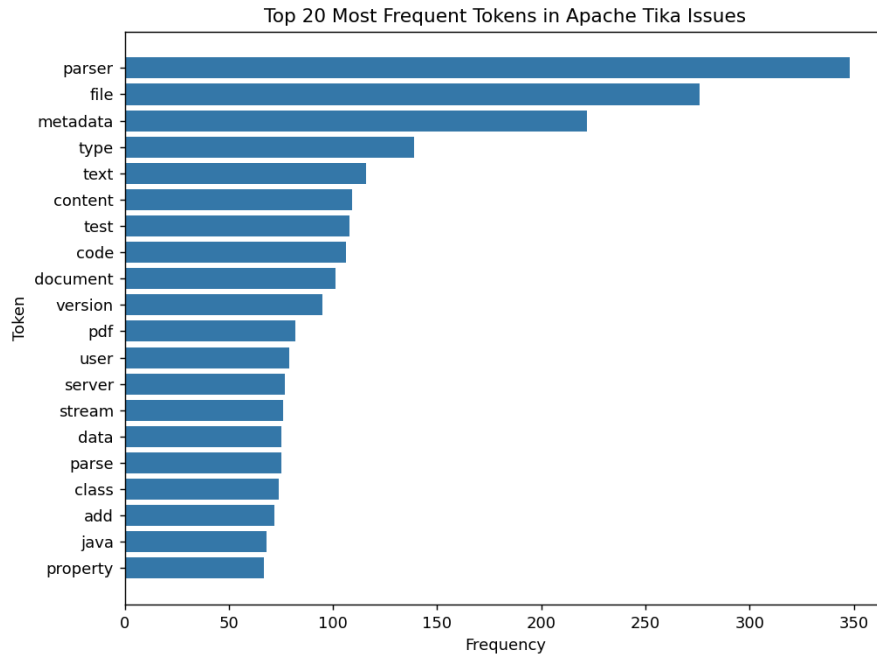


Figure 3.1: Top 20 most frequent tokens in the preprocessed Apache Tika issues.

3.2 Week 2: LDA Topic Modeling

Iteration 1: 10 topics

LDA was trained on the 211 preprocessed issues with 10 topics. Table 3.1 lists the top three terms, the number of assigned issues and a brief theme for each topic. Topic 8 was the largest group with 51 issues, covering the core parser API. Topic 4 captured dependency-vulnerability reports (Sonatype, jar files, CVE). Most issues received a dominant-topic probability above 0.99, reflecting the sparse distribution produced by the low α .

Table 3.1: Iteration 1: top terms per topic (10 topics, $\alpha = \beta = 0.01$).

Topic	Top 3 terms	Issues	Theme
0	metadata, code, name	22	Metadata and EXIF
1	file, text, parser	21	File and content parsing
2	image, embedded, content	13	Image and OCR
3	metadata, mapping, java	11	Metadata mapping
4	file, sonatype, java	26	Dependency vulnerabilities
5	document, file, parser	18	Document parsing
6	test, pdf, file	16	PDF testing
7	log, file, version	19	Logging and CVE
8	parser, metadata, content	51	Parser API
9	parser, file, content	14	Content and configuration

Coherence analysis

C_V coherence scores were computed for $k = 1$ to $k = 10$ (Table 3.2, Figure 3.2). The score peaked at $k = 7$ ($C_V = 0.3462$), which was used for Iteration 2.

Table 3.2: C_V coherence scores for $k = 1$ to $k = 10$.

k	C_V coherence
1	0.2228
2	0.2270
3	0.2539
4	0.2611
5	0.3156
6	0.3420
7	0.3462
8	0.3299
9	0.3268
10	0.3313

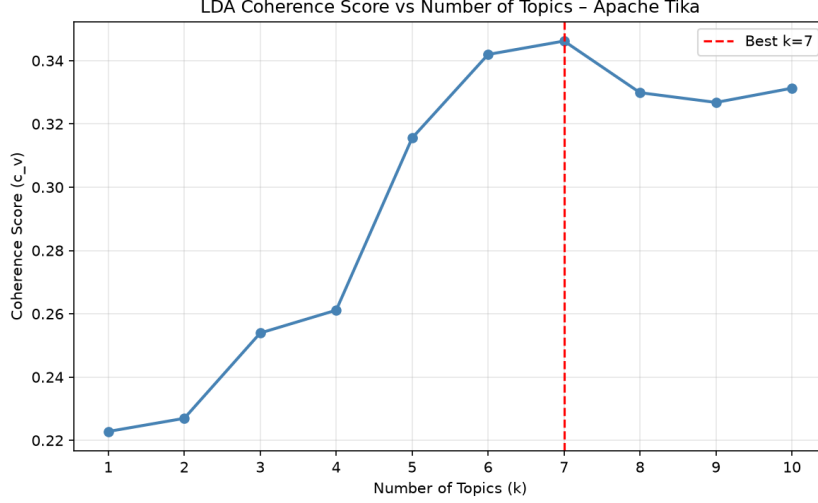


Figure 3.2: C_V coherence score for $k = 1$ to $k = 10$. The highest value is at $k = 7$.

Iteration 2: ontology replacement

Before re-running LDA, tokens that appeared in the ontology were replaced with their class labels. A total of 2,832 replacements were made across all 211 issues. LDA was then trained with $k = 7$ (Table 3.3). The class label **COMPONENT** appeared as a top term in all seven topics, reflecting that most issues involve software components. Topic 4 was the largest (89 issues), covering the general parser framework; Topic 3 was the most distinct, capturing security vulnerabilities (*vulnerability*, *cve*, *sonatype*).

Table 3.3: Iteration 2: top terms per topic with ontology replacement ($k = 7$).

Topic	Top 3 terms	Issues
0	TECHNOLOGY, DATA, COMPONENT	17
1	COMPONENT, TECHNOLOGY, DATA	63
2	property, log, COMPONENT	9
3	COMPONENT, version, vulnerability	21
4	COMPONENT, parser, CONNECTOR	89
5	test, pdf, COMPONENT	4
6	memory, text, DATA	8

3.3 Week 2: BERTopic Topic Modeling

BERTopic was run on the 211 preprocessed issues using **all-MiniLM-L6-v2** sentence embeddings. The default HDBSCAN clustering collapsed the corpus into only two topics (178 and 33 issues), too coarse to compare with LDA or to test co-occurrences. We therefore replaced the clustering step with KMeans ($k = 7$), yielding seven balanced, interpretable topics with every issue assigned and no outliers (Table 3.4); a fixed random state of 42 makes the run reproducible. Compared with LDA, BERTopic produced a more even distribution (17–44 issues per topic versus LDA’s 4–89) and kept the vulnerability topic (Topic 2) sharply separated.

Table 3.4: BERTopic topics ($k = 7$): top terms, number of issues, and theme.

Topic	Top terms	Issues	Theme
0	test, pdf, type, image, file	44	File-type detection / PDF & image
1	metadata, property, field, parser	35	Metadata extraction
2	version, vulnerability, log, jar	33	Dependency & security vulnerabilities
3	parser, metadata, document, file	31	Document parsing & configuration
4	parser, java, microsoft, parse	29	Office / language-specific parsers
5	file, server, url, fetcher	22	Tika-server & fetchers
6	process, stream, input, processor	17	Streaming / async processing

3.4 RQ1: Topics and Keywords from LDA and BERTopic

What topics emerge from LDA and BERTopic and what are the common keywords per topic? Both algorithms surface the same broad areas of Apache Tika: a large parser/metadata core, document/PDF/image handling, the tika-server, and a distinct security and dependency-vulnerability topic (LDA Topic 3; BERTopic Topic 2). LDA (after ontology replacement) is dominated by one large topic (Topic 4, 89 issues) and relies heavily on the abstract ontology labels, whereas BERTopic gives a more balanced and more concrete keyword picture (e.g. `microsoft`, `fetcher`, `image`, `vulnerability`). The number of issues per topic is reported in Tables 3.3 and 3.4.

3.5 Week 3: Characteristics and Co-occurrences

3.5.1 RQ2: Characteristics of the issues in each topic

For each topic we compared four numeric characteristics—number of comments, number of attachments, description length (characters) and preprocessed token count—using Kruskal–Wallis omnibus tests (the count fields are strongly right-skewed), followed by Dunn post-hoc tests where the omnibus test was significant. Issue type was compared with a χ^2 test.

Table 3.5: Kruskal–Wallis omnibus tests for issue characteristics per topic.

Model	Characteristic	H	p	Sig.
LDA	comments	2.35	0.885	no
LDA	attachments	2.76	0.838	no
LDA	description length	4.58	0.599	no
LDA	token count	4.71	0.582	no
BERTopic	comments	7.09	0.312	no
BERTopic	attachments	35.35	< 0.001	yes
BERTopic	description length	14.65	0.023	yes
BERTopic	token count	13.81	0.032	yes

LDA topics do *not* separate issues by any metadata characteristic (all $p > 0.27$). BERTopic topics do: they differ significantly in attachments, description length and token count, though not in comments. The vulnerability topic (BERTopic Topic 2) has the longest, most detailed descriptions. Issue type is also strongly associated with BERTopic topic ($\chi^2 = 164$, $p < 0.001$; Figure 3.3). We note that 67% of the expected cell counts here are below five, so this χ^2 is approximate and should be read together with the heatmap.

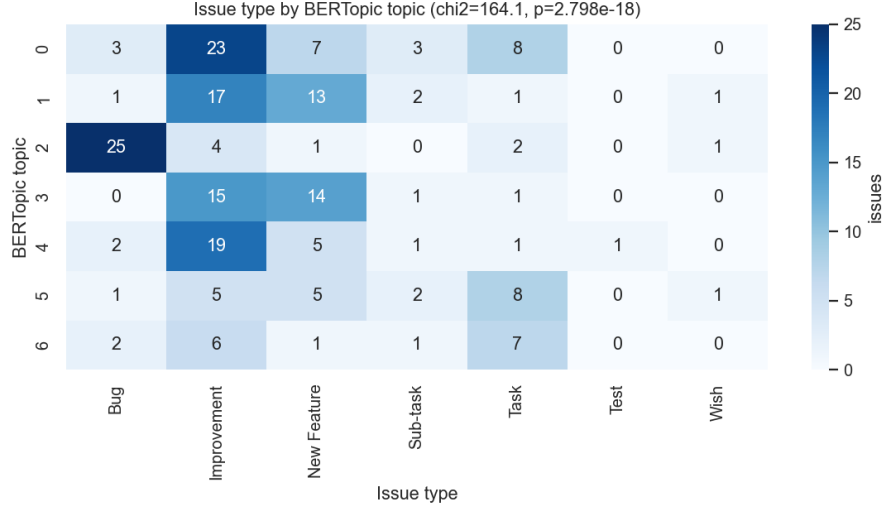


Figure 3.3: Issue type by BERTopic topic.

3.5.2 RQ3: Co-occurrences

(a) LDA topics versus BERTTopics

The 7×7 contingency table of LDA versus BERT topic assignments is significantly associated ($\chi^2 = 114.9$, $\text{dof} = 36$, $p = 3.6 \times 10^{-10}$). Standardized residuals identify six topic pairs that co-occur more or less than chance. The strongest is **LDA Topic 3** $\leftrightarrow$ **BERT Topic 2** (residual = 5.9): both are the security/vulnerability topic, i.e. the two models independently isolate the same issues (Figure 3.4). As above, 71% of expected cells are below five, so the omnibus χ^2 is approximate and the residual pattern is reported as descriptive evidence of alignment.

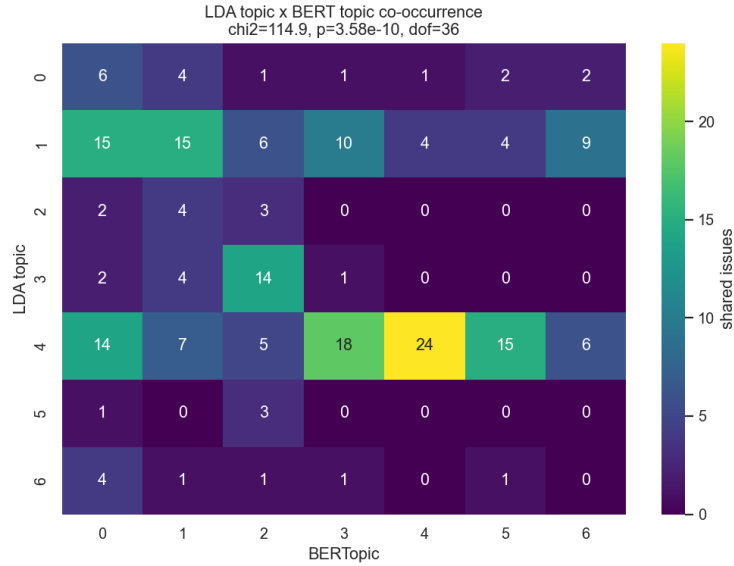


Figure 3.4: Issues shared between each LDA topic (rows) and BERTopic (columns).

(b) Topics versus the three design-decision types

Each issue belongs to one topic but may carry several decision types, so for every (topic, decision) pair we ran a Fisher exact test (odds ratio) and corrected p -values within each model with Benjamini–Hochberg (Table 3.6). The security topic (BERTopic Topic 2 / LDA Topic 3) is

overwhelmingly associated with **property** and **executive** decisions and essentially never an **existence** decision: dependency upgrades and vulnerability fixes change *how* something is done or *which* technology is used, not *whether* a component exists. Conversely, the metadata and document-parsing topics (BERTopic 1, 3) and the parser-core topic (LDA 4) are associated with **existence** decisions (Figure 3.5).

Table 3.6: Significant (FDR < 0.05) topic $\leftrightarrow$ decision associations (selected).

Model	Topic	Decision	% w/ dec.	OR	p_{adj}
BERTopic	2 (security)	property	97.0	66.2	< 0.001
BERTopic	2 (security)	executive	90.9	34.5	< 0.001
BERTopic	2 (security)	existence	6.1	0.02	< 0.001
BERTopic	1 (metadata)	existence	88.6	4.43	0.009
BERTopic	3 (doc parsing)	existence	90.3	5.28	0.009
LDA	3 (security)	executive	76.2	8.06	< 0.001
LDA	3 (security)	property	71.4	3.83	0.048
LDA	4 (parser core)	existence	78.7	2.47	0.032

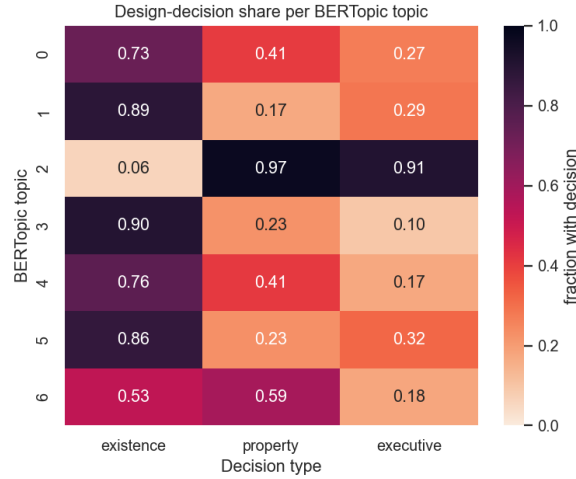


Figure 3.5: Fraction of each BERTopic topic carrying each design-decision type.

(c) Co-occurrence among the three decision types

We also tested whether the decision types co-occur with each other (Table 3.7). **Existence** decisions are largely mutually exclusive with **property** ($\phi = -0.80$; because every issue has one or the other, the odds ratio collapses to ≈ 0) and, less strongly, with **executive** ($\phi = -0.38$). In contrast, **property** and **executive** decisions tend to co-occur ($\phi = +0.27$, OR = 3.2). Deciding *whether* a component exists is a different kind of decision from deciding *how* it should behave or *which* technology to use, and the latter two often travel together (Figure 3.6). Every issue carries at least one decision type; existence-only is the largest group (44.5%) and property+executive is the most common combination (19.0%).

Table 3.7: Co-occurrence among the three design-decision types.

Pair	Both	OR	ϕ	p_{adj}	Association
existence $\leftrightarrow$ property	22	≈ 0	-0.80	< 0.001	strong negative
existence $\leftrightarrow$ executive	30	0.19	-0.38	< 0.001	negative
property $\leftrightarrow$ executive	43	3.19	+0.27	< 0.001	positive

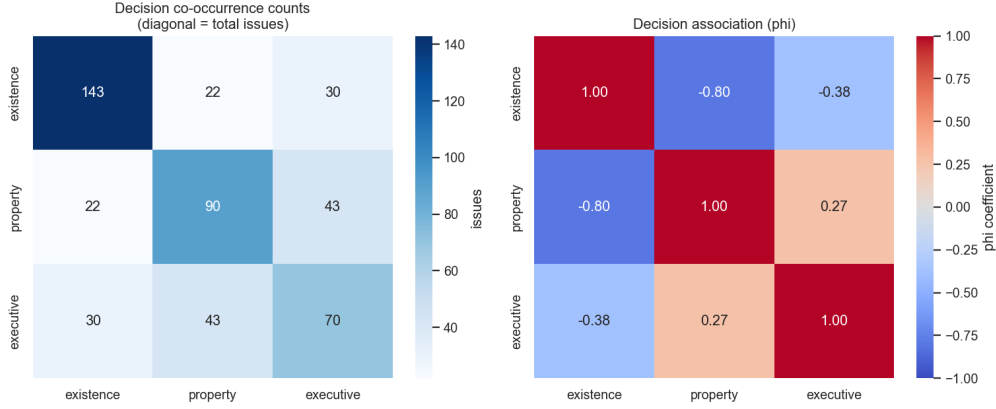


Figure 3.6: Left: co-occurrence counts among decision types (diagonal = total issues). Right: phi association between decision types.

Chapter 4

Discussion

The Week 1 preparation was successful: all 211 issues were downloaded and still had usable text after preprocessing. The most frequent tokens (`parser`, `file`, `metadata`, `document`, `pdf`, `content`) reflect Tika’s role in parsing files and extracting text and metadata, and terms such as `vulnerability`, `cve` and `vulnerable` already hinted at a security topic.

The LDA results confirmed this. Iteration 1 surfaced two security-related topics (dependency vulnerabilities and logging/CVE), and the coherence analysis rose steadily to a peak at $k = 7$, indicating that seven is a meaningful granularity. Ontology replacement in Iteration 2 collapsed many tokens into six class labels; this surfaced the architectural dimension (the dominance of `COMPONENT`) but also reduced the discriminative power between topics, leaving Topics 0, 1 and 6 harder to interpret.

BERTopic was more informative once re-tuned. The original two-topic run was too coarse; the seven-topic KMeans configuration produced balanced, concrete topics and, unlike LDA, its topics differed significantly in issue characteristics (RQ2) — particularly attachments and description length — and aligned strongly with LDA on the security topic (RQ3a).

The headline finding (RQ3b) is the sharp division of labour between decision types and topics: the security topic maps onto *property* and *executive* decisions, while component-adding topics map onto *existence*. RQ3c reinforces this at the decision level: existence decisions are almost mutually exclusive with property, while property and executive travel together. Together these results suggest that topic membership is a useful predictor of the *type* of design decision an issue records.

Chapter 5

Threats to Validity

5.1 Internal Validity

Preprocessing choices (stop-word removal, lemmatization, deletion of short tokens) affect the vocabulary and therefore the topics. A fixed random seed makes the models reproducible, but different hyperparameters for α , β or the number of passes could change topic assignments. The coherence optimisation was limited to $k = 1$ to $k = 10$, so a globally better k could lie outside this range, and C_V measures statistical co-occurrence of terms rather than whether topics are meaningful for design decisions.

5.2 External Validity

The study focuses only on Apache Tika, so results may not generalise to other projects, each of which has its own terminology and issue discussions.

5.3 Construct Validity

Only the issue summary and description were used for text analysis; useful information in comments or attachments was not included. BERTopic used KMeans ($k = 7$) because HDBSCAN under-clustered this small corpus; KMeans forces a fixed k and assigns every issue, which is convenient for co-occurrence but removes BERTopic’s automatic topic-count selection. Several χ^2 tests have more than 20% of expected cells below five (flagged as `pct_expected_lt5` in the output CSVs), so the pairwise conclusions rely on exact Fisher tests, and Benjamini–Hochberg correction was applied across all topic $\times$ decision pairs.

Chapter 6

Conclusion and Future Work

Across the three weeks we prepared the Apache Tika issue dataset, applied two topic models, and analysed the resulting topics. Week 1 produced a clean corpus, vocabulary and document-term matrix. Week 2 ran LDA (with a coherence-selected $k = 7$ and an ontology-enriched iteration) and BERTopic, answering RQ1: both models recover a parser/metadata core, document and image handling, a tika-server area, and a distinct security topic. Week 3 answered RQ2 and RQ3: BERTopic topics differ significantly in issue characteristics, topics from the two models align on the security theme, and topics co-occur meaningfully with the three decision types—security with property/executive, component work with existence.

Future work could extend the coherence search beyond $k = 10$, incorporate issue comments into the text, refine the ontology so LDA Iteration 2 stays interpretable, and validate the topic–decision associations on a second Apache project to test generalisability.

Appendix A

Team Effort and Contributions

Tables A.1–A.3 summarise the main activities per week. Rows marked *(to be filled)* are to be completed by the respective team members.

Table A.1: Summary of team effort in Week 1.

Name	Hours	Activities
Obinna	6	Organised the Week 1 folder; processed the Tika issue list and downloaded the Jira issue data; preprocessed summaries and descriptions; created the vocabulary and document-term matrix.
Anil	8	Fetches all issues from the Jira API; applied tokenisation, stop-word removal, lemmatisation and ontology-class replacement; created the filtered vocabulary.
Brian	–	<i>(to be filled)</i>
Namit	0	No contribution in Week 1.

Table A.2: Summary of team effort in Week 2.

Name	Hours	Activities
Brian	7	Ran the topic-modeling algorithms (LDA and BERTopic); wrote the Week 2 LDA report section.
Namit	2	Fixed the BERTopic results: the initial run collapsed into two topics, so re-ran BERTopic with KMeans ($k = 7$) to obtain seven balanced, interpretable topics and the per-issue assignments used for the Week-3 co-occurrence analysis.
Obinna	–	<i>(to be filled)</i>
Anil	–	<i>(to be filled)</i>

Table A.3: Summary of team effort in Week 3.

Name	Hours	Activities
Namit	9	RQ2: per-topic characteristics (comments, attachments, description length, tokens) with Kruskal-Wallis and Dunn tests, box plots and issue-type χ^2 . RQ3: LDA $\times$ BERT co-occurrence, topic $\times$ decision Fisher tests (FDR) and decision-type co-occurrence. Wrote and integrated the Week-3 report sections.
Brian	–	<i>(to be filled)</i>
Obinna	–	<i>(to be filled)</i>
Anil	–	<i>(to be filled)</i>

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